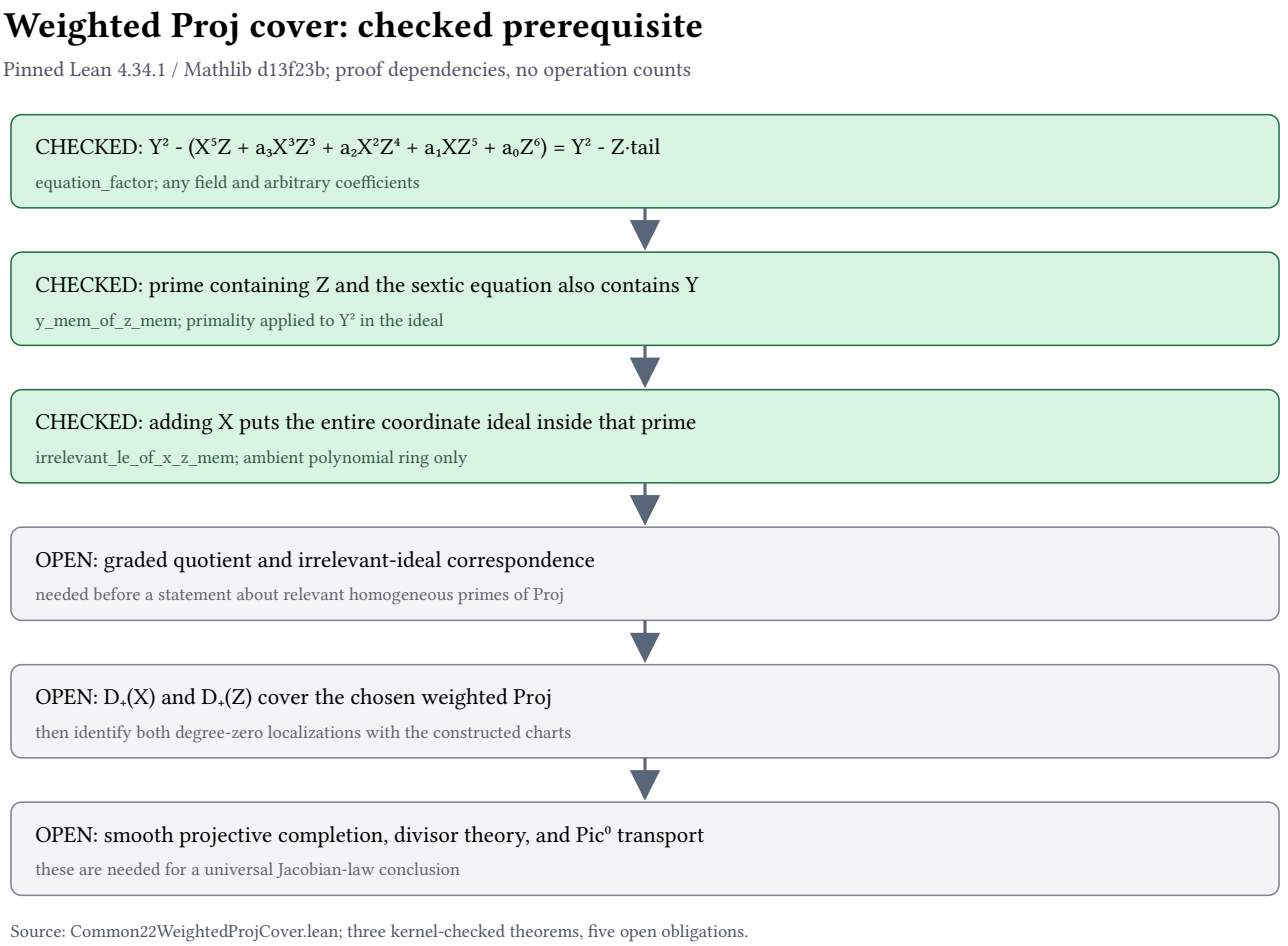
Weighted Proj cover: checked prime-containment prerequisite

Pinned Lean 4.34.1, Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612`. The parameter field and the four quintic coefficients are arbitrary. This is a formal proof-status diagram, with no measured operation counts or new cost ratio.



The checked factorization is `Y^2 - Z * tail`. For any prime ideal in the ambient three-variable polynomial ring containing `Z` and the equation, primality implies that it contains `Y`; if it also contains `X`, the ideal generated by `X,Y,Z` lies inside the prime. The Lean compiler exited 0 and all three theorem audits list only `propext`, `Classical.choice`, and `Quot.sound`.

This does **not** construct the graded sextic quotient, identify its irrelevant ideal, or state a `Proj` open-cover theorem. The exact next bridge is to prove those graded-ring steps, identify both degree-zero localizations with the existing two chart rings, and establish smooth projective and `Pic^0` compatibility. The sibling CSV retains each checked and open obligation.

Proof source: `Common22WeightedProjCover.lean`. Compiler receipt: `weighted-proj-cover-build.log`. Native identity certificate: `weighted-proj-cover-identity.json` (the polynomial factorization only).